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UNIVERSITI TUN HUSSEIN ONN MALAYSIA

**FINAL EXAMINATION
SEMESTER I
SESSION 2024/2025**

COURSE NAME	:	COMMUNICATION AND NETWORK SECURITY
COURSE CODE	:	BIS 30903
PROGRAMME CODE	:	BIS
EXAMINATION DATE	:	FEBRUARY 2025
DURATION	:	3 HOURS
INSTRUCTION	:	1. ANSWER ALL QUESTIONS 2. THIS FINAL EXAMINATION IS CONDUCTED VIA ☐ Open book ☒ Closed book 3. STUDENTS ARE PROHIBITED TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK

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THIS QUESTION PAPER CONSISTS OF **TEN (10)** PAGES

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PART A

Choose the **BEST** answer.

Q1 What is the primary purpose of Wifi Protected Access 3 (WPA3) in wireless networks?

- (a) To improve the speed of wireless connections.
- (b) To provide stronger encryption for wireless communications.
- (c) To allow more devices to connect simultaneously.
- (d) To eliminate the need for passwords.

(2 marks)

Q2 Which of the following actions can mitigate risks associated with open Wi-Fi networks?

- (a) Using public Domain Name Service (DNS) servers.
- (b) Implementing a Virtual Private Network (VPN).
- (c) Disabling firewalls.
- (d) Sharing passwords publicly.

(2 marks)

Q3 A company wants to allow only web traffic to enter its internal network. Which type of firewall rule should be implemented?

- (a) Allow all traffic.
- (b) Deny all traffic except HTTP/HTTPS.
- (c) Allow all incoming connections.
- (d) Block traffic from all IP addresses.

(2 marks)

Q4 Which of the following describes a stateful firewall?

- (a) It filters packets based on static rules only.
- (b) It monitors the state of active connections and makes decisions based on the context of traffic.
- (c) It allows all traffic until a rule is violated.
- (d) It does not maintain any connection information.

(2 marks)

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- Q5** What is the primary goal of deploying a honeypot in a network?
- (a) To speed up network performance.
 - (b) To gather intelligence on attackers and their methods.
 - (c) To block malicious traffic.
 - (d) To enhance user experience.
- (2 marks)
-
- Q6** When analyzing the effectiveness of a honeypot, which metric is most relevant?
- (a) Number of legitimate users.
 - (b) Amount of malware detected and analyzed.
 - (c) Total bandwidth consumed.
 - (d) User satisfaction ratings.
- (2 marks)
-
- Q7** How does an Intrusion Detection System (IDS) primarily differ from an Intrusion Prevention System (IPS)?
- (a) An IDS prevents intrusions; an IPS only detects them.
 - (b) An IDS analyzes traffic; an IPS analyzes and acts on that traffic.
 - (c) An IDS is more costly than an IPS.
 - (d) An IDS does not require network configuration.
- (2 marks)
-
- Q8** In the context of an IPS, what does "anomaly detection" refer to?
- (a) Identifying known malicious signatures.
 - (b) Monitoring network traffic to identify abnormal behavior.
 - (c) Blocking all unauthorized access attempts.
 - (d) Filtering out spam emails.
- (2 marks)

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- Q9** What type of access control ensures that users can only access resources based on their roles within an organization?
- (a) Discretionary Access Control (DAC).
 - (b) Mandatory Access Control (MAC).
 - (c) Role-Based Access Control (RBAC).
 - (d) Rule-Based Access Control (RBAC).
- (2 marks)
-
- Q10** When analyzing access control logs, which of the following could indicate a potential security breach?
- (a) Successful login attempts during business hours.
 - (b) Multiple failed login attempts from the same IP address.
 - (c) Regular password changes.
 - (d) Access requests from approved locations.
- (2 marks)
-
- Q11** Which practice is essential for maintaining the security of network devices?
- (a) Regularly updating firmware and software.
 - (b) Leaving default passwords unchanged.
 - (c) Limiting access to devices to only local users.
 - (d) Disabling logging features.
- (2 marks)
-
- Q12** In assessing the security of a router, which feature should be enabled to protect against unauthorized access?
- (a) Guest network.
 - (b) Wi-Fi Protected Setup (WPS).
 - (c) Strong administrative passwords.
 - (d) Universal Plug and Play (UPnP).
- (2 marks)

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- Q13** Why is it critical to regularly update network security software?
- (a) To increase network bandwidth.
 - (b) To ensure the software is compatible with new hardware.
 - (c) To patch vulnerabilities and enhance security features.
 - (d) To make the software easier to use.
- (2 marks)
-
- Q14** When analyzing the impact of software updates, which factor is most important?
- (a) The number of features added.
 - (b) The security vulnerabilities addressed in the update.
 - (c) The marketing strategy behind the software.
 - (d) User interface changes.
- (2 marks)
-
- Q15** What is a common characteristic of phishing emails?
- (a) They contain no links or attachments.
 - (b) They often appear to be from trusted sources.
 - (c) They always use formal language.
 - (d) They are usually sent from company emails.
- (2 marks)
-
- Q16** Which technique is most effective in analyzing the risk of social engineering attacks within an organization?
- (a) Training employees to recognize phishing attempts.
 - (b) Increasing the number of firewalls.
 - (c) Implementing stronger password policies.
 - (d) Reducing the number of network devices.
- (2 marks)
-
- Q17** What is the primary goal of a security policy in an organization?
- (a) To provide a framework for compliance with regulations.
 - (b) To outline the technical specifications of security tools.
 - (c) To establish a budget for IT spending.
 - (d) To increase the number of employees in IT.

(2 marks)

- Q18** In a defense-in-depth strategy, which layer is considered the last line of defense against attacks?
- (a) Perimeter security (firewalls).
 - (b) Endpoint security (antivirus).
 - (c) Application security.
 - (d) User education and awareness.

(2 marks)

- Q19** What is Address Resolution Protocol (ARP) spoofing primarily used for?
- (a) To increase network speed.
 - (b) To redirect traffic and potentially capture sensitive information.
 - (c) To prevent Distributed Denial of Service (DDoS) attacks.
 - (d) To enhance wireless security.

(2 marks)

- Q20** During a man-in-the-middle (MITM) attack, what is typically compromised?
- (a) The integrity of the data only.
 - (b) The confidentiality and integrity of the communication between two parties.
 - (c) The speed of the network connection.
 - (d) The physical security of the network devices.

(2 marks)

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PART B

Q21 Based on information provided in **Figure Q21.1** and **Table Q21.1**, write appropriate command(s) to do the following task.

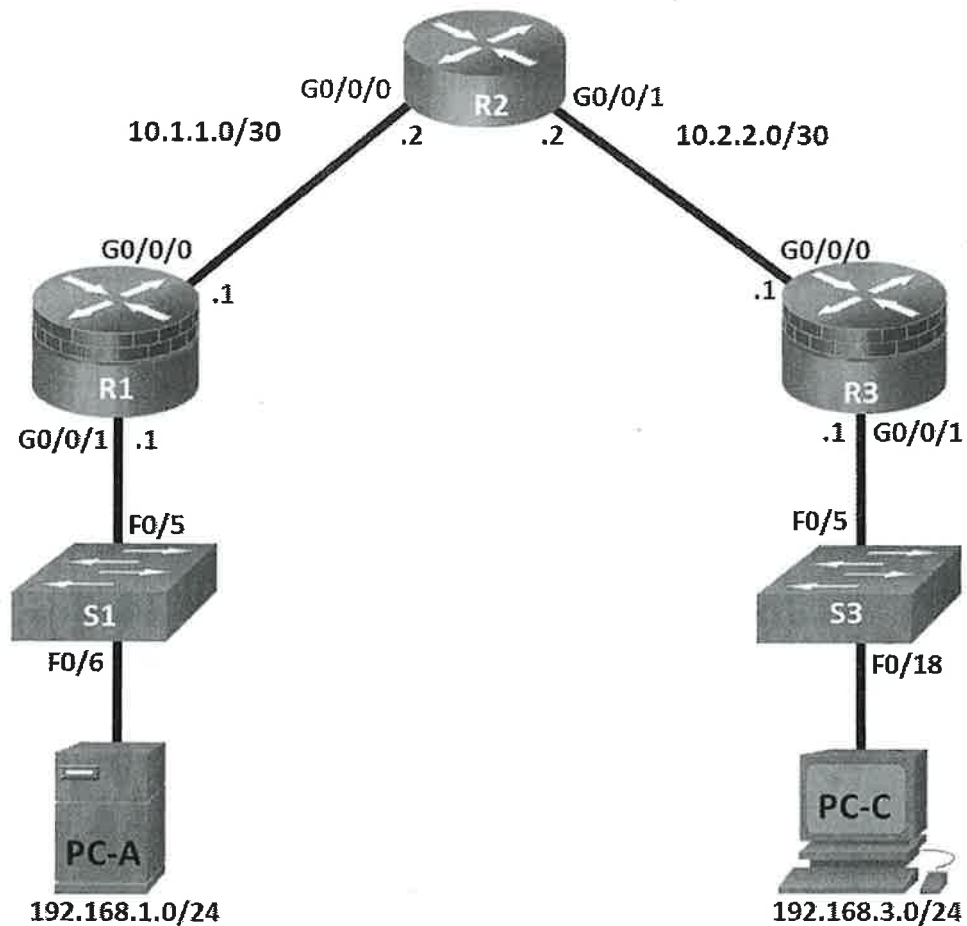


Figure Q21.1: Topology of a Network

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Table Q21.1: Device and Interface Address Assignments

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
	G0/0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
R2	G0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	G0/0/1	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/0/0	10.2.2.1	255.255.255.252	N/A	N/A
	G0/0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

- (a) Configure basic settings such as host name, interface IP addresses, and access passwords. (3 marks)
- (b) Configure static routing. (1 mark)
- (c) Configure a local database user and local access for the console line. (1 mark)
- (e) Configure domain name (1 mark)
- (f) Configure encryption key (1 mark)
- (g) Enable SSH on vty (1 mark)
- (h) Configure the local user database using Cisco IOS. (1 mark)
- (i) Configure AAA local authentication using Cisco IOS. (1 mark)

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- Q22** A network engineer of a large company decided to install a firewall to protect its operation. The company has a web server, a HTTP proxy server, two internally accessible corporate servers, an email server, DNS, VoIP Server and DHCP server. There are 2000 computer accessing the internal and external resources.

Based on given scenario, answer the following questions.

- (a) Describe **FIVE (5)** types of firewall. (10 marks)
- (b) Draw **TWO (2)** possible firewall designs the engineer may use to protect this company. Please labels your diagrams. (5 marks)
- (c) Explain the strength and weaknesses of each design in **Q22(b)**. (8 marks)

- Q23** The company's IT department has detected unauthorized wireless access points within the corporate network. These rogue APs have raised concerns about network security, prompting the need for a thorough understanding of rogue APs and strategies for addressing this issue.

Based on given scenario, answer the following questions.

- (a) Who and why use Rouge Access Points? (4 marks)
- (b) Recommend **FOUR (4)** mitigation techniques to address the threat of rogue access points and enhance the security of the company's wireless LAN network. (8 marks)

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- Q24** A large financial institution, JimBank has implemented both Intrusion Prevention Systems (IPS) and Intrusion Detection Systems (IDS) to protect its network. An alert is triggered, indicating a potential security threat.

Differentiate between the roles of an IPS and an IDS in responding to this alert.

(5 marks)

Q25 AntiLolos, a medium-sized e-commerce company experiences a sudden surge in traffic to its website, causing performance issues. The IT team suspects a Distributed Denial of Service (DDoS) attack.

Propose TWO(2) mitigations to address the given threats.

(5 marks)

- END OF QUESTIONS -

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